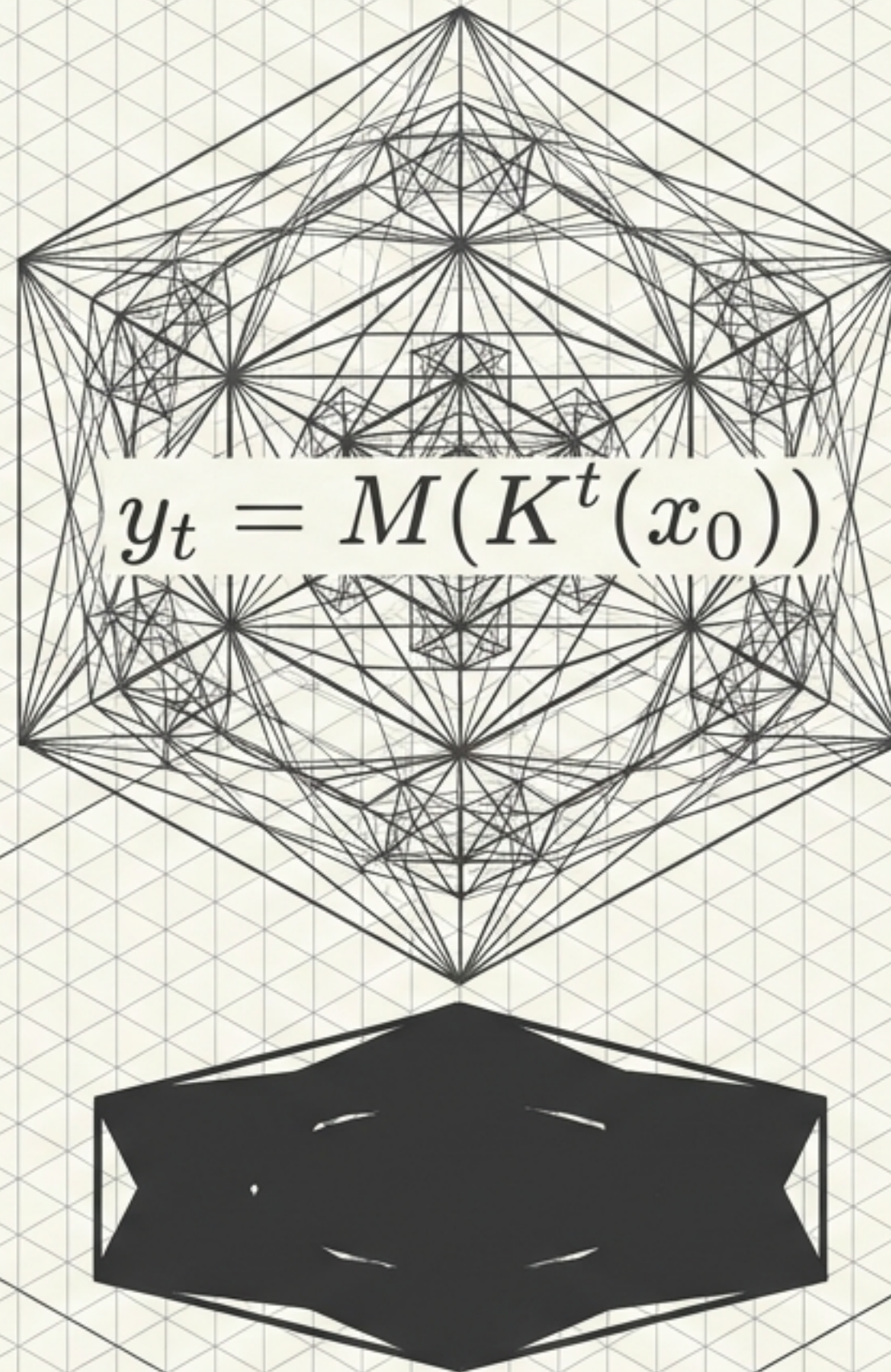




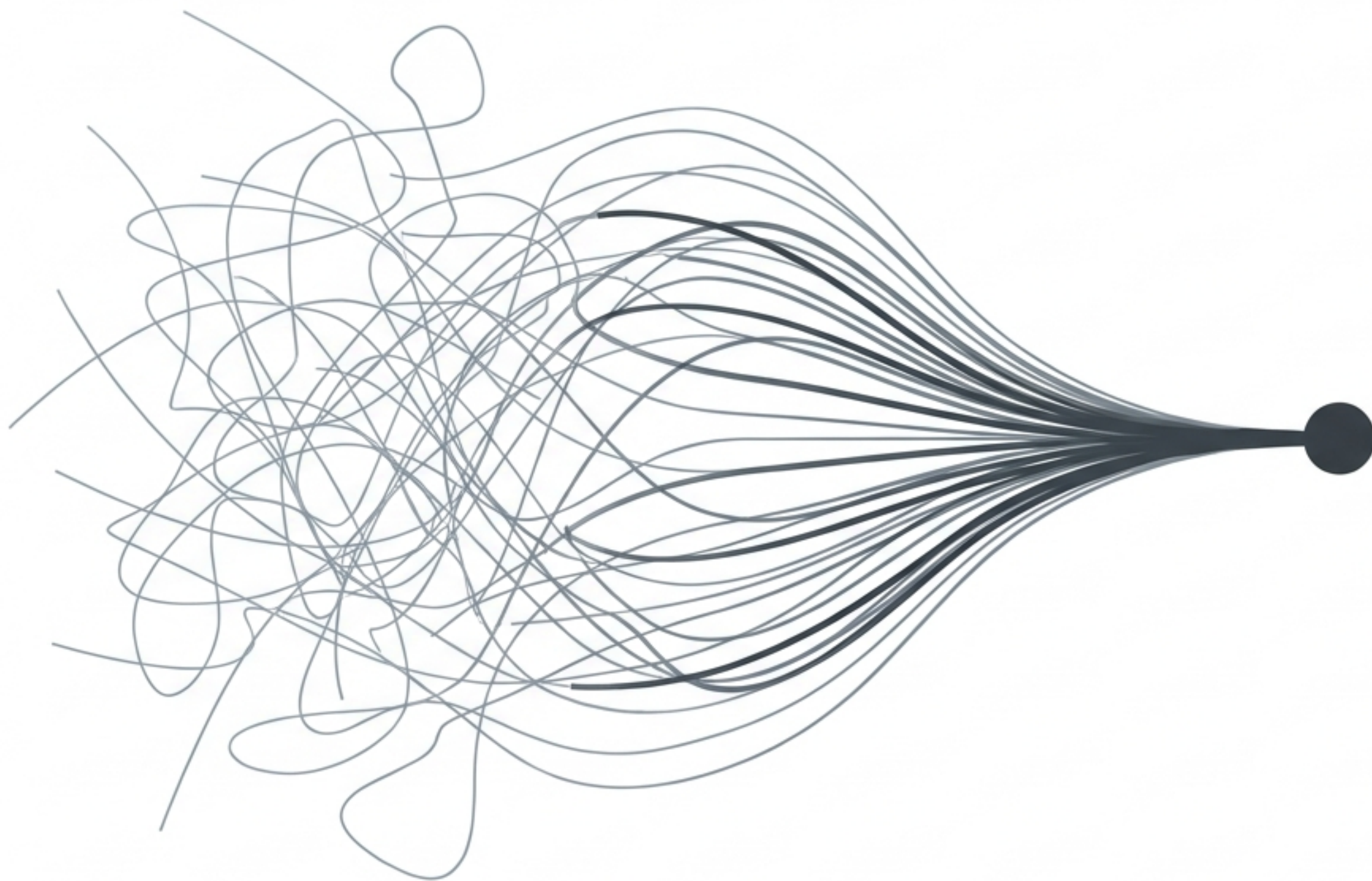
Title: **THE SHAPE OF SEARCH**

Subtitle: Reachability, Measurement, and Attraction in  
Physical, Cognitive, and Conversational Systems.

Author: Flyxion Independent Researcher



# The Endpoint Fallacy



CORE AXIOM:  
**OBSERVED  
TRAJECTORY  $\neq$   
UNDERLYING  
DYNAMICS**

We judge optimizers by minima, minds by capacity parameters, and agents by consensus.

But an endpoint description omits the conditions of its own intelligibility. It tells us where a system is, but erases how it arrived and where it can go.

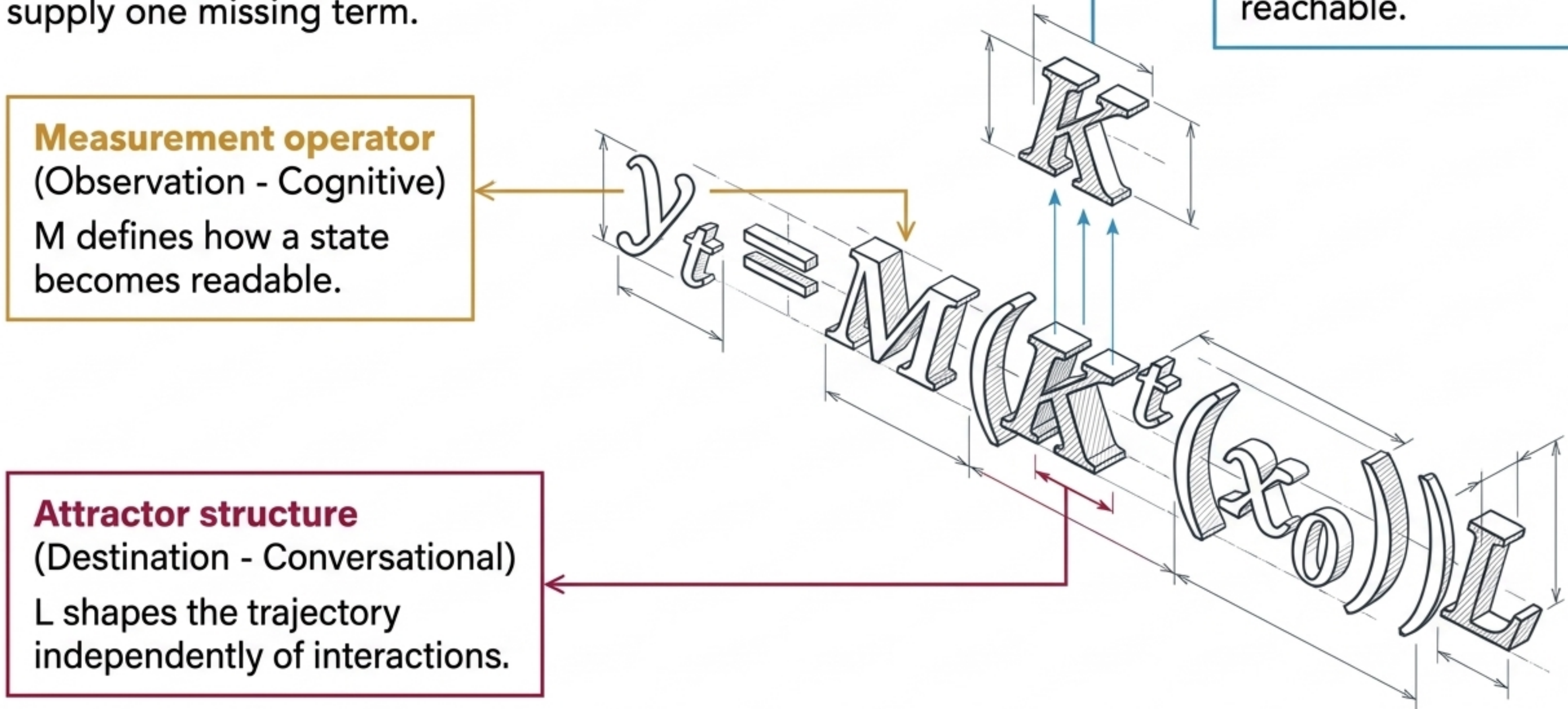


# The Three Missing Terms

Endpoints omit three structures. Three recent empirical results across isolated domains each supply one missing term.

**Measurement operator**  
(Observation - Cognitive)  
M defines how a state becomes readable.

**Transition dynamics**  
(Reachability - Physical)  
K determines what is reachable.

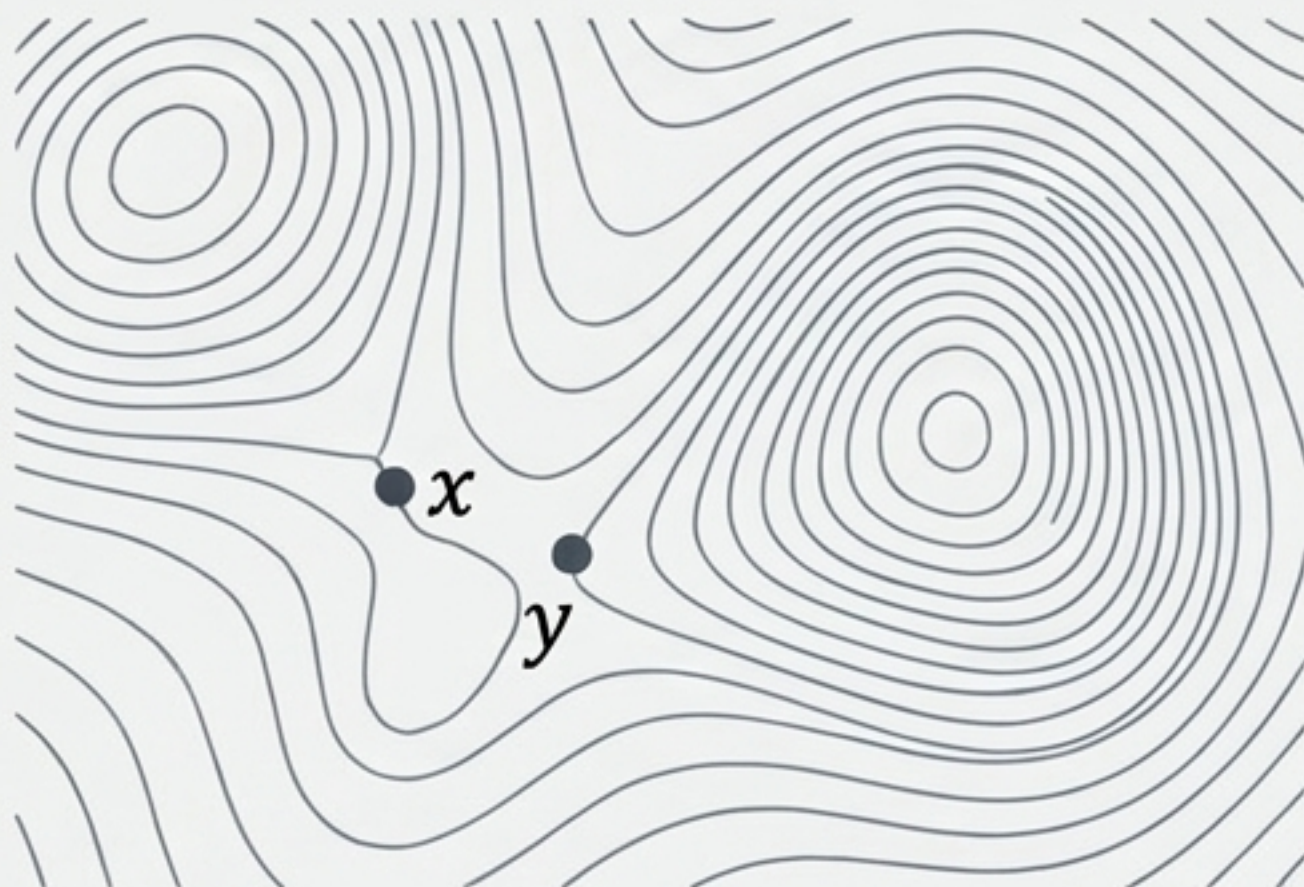




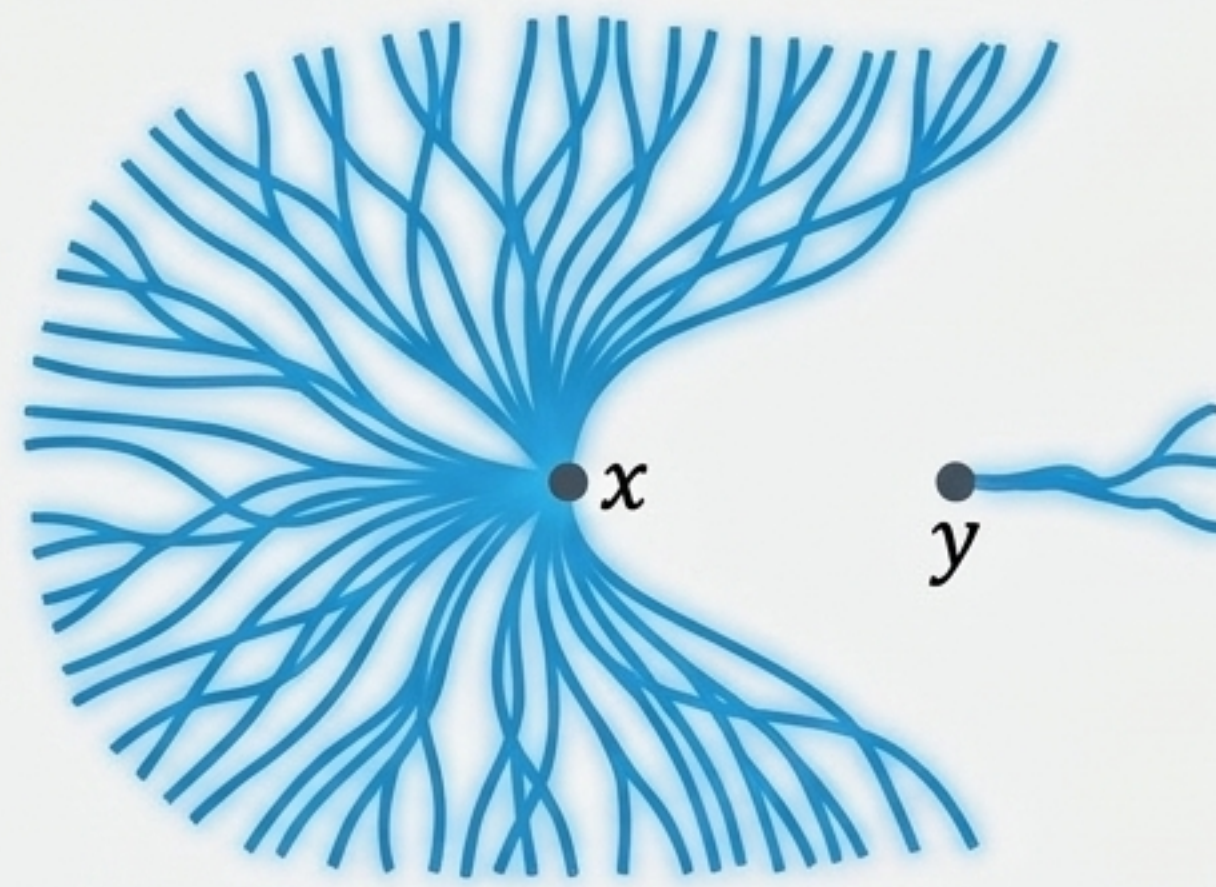
# Case 1: Search is Not Enumeration

$$E(\mathbf{x}) = E(\mathbf{y}) \neq \Gamma(\mathbf{x}) \cong \Gamma(\mathbf{y})$$

**Energy Geometry**



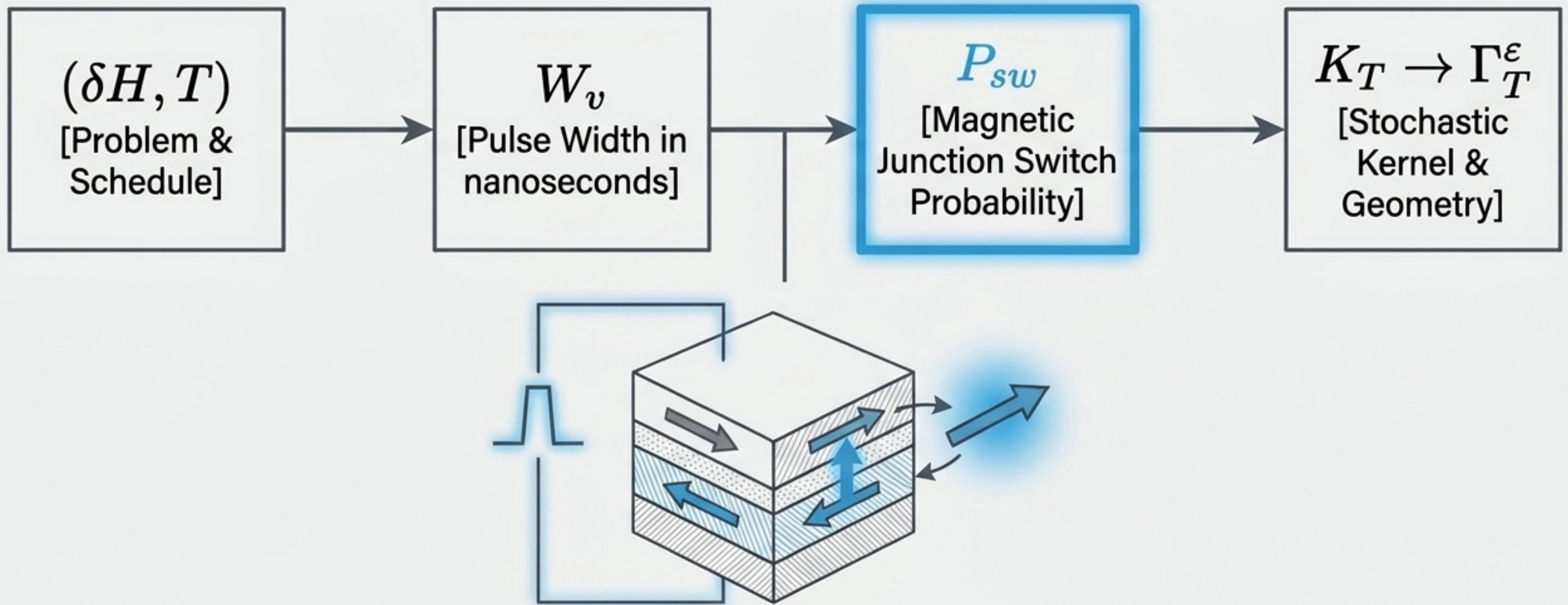
**Continuation Geometry**



Two configurations can share the exact same energy level but differ entirely in what they can become. The transition kernel  $K$  dictates the search geometry, not just the objective function.



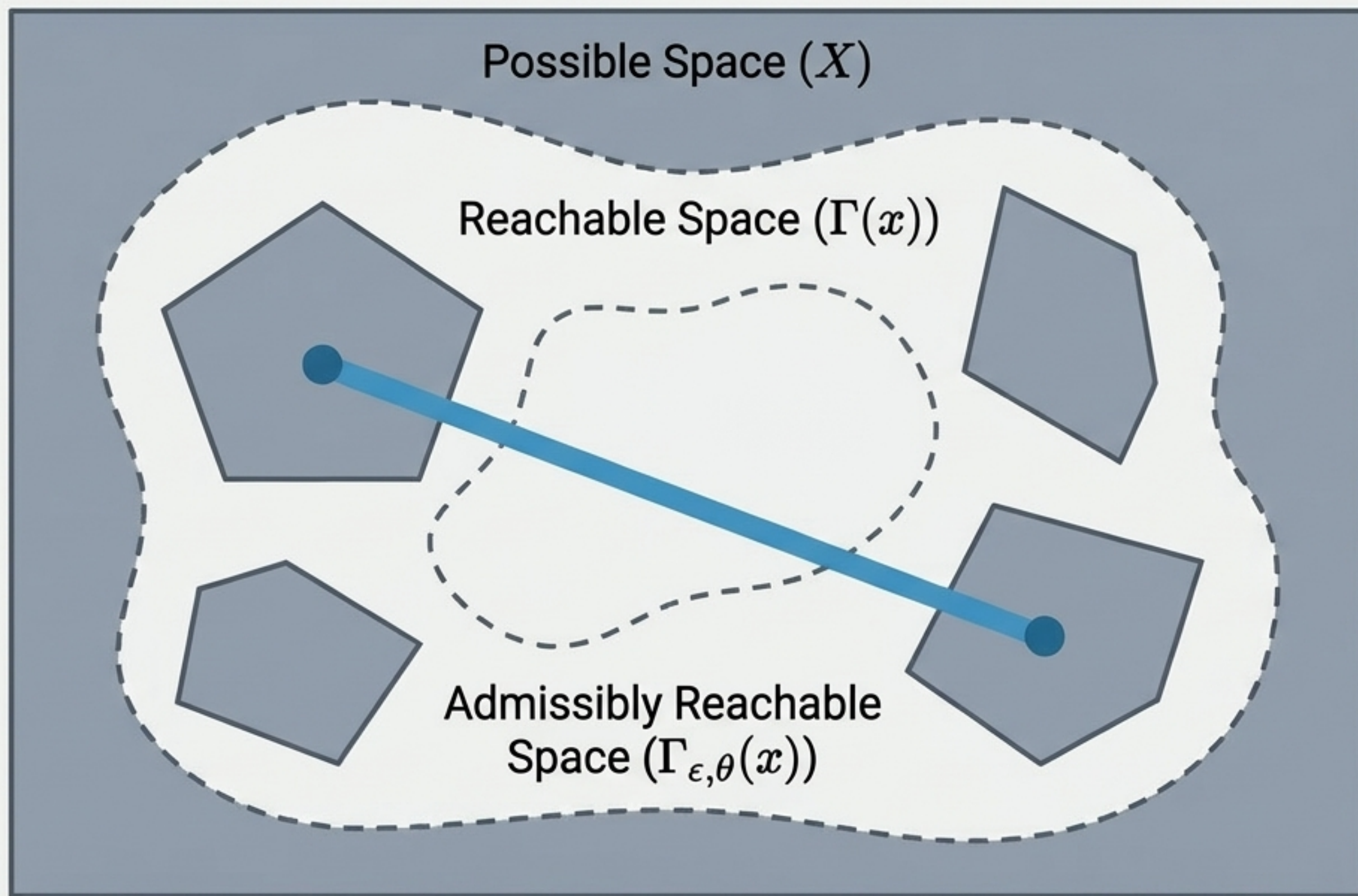
# Probability as a Material Property



In the VC-MRAM Ising machine, chance is not assembled from digital bits. The problem, the schedule, and the matter meet in one scalar. A 0.8ns pulse does not store a bit; it sets a probability measure.



# Admissibility & The Geometry of Constraints



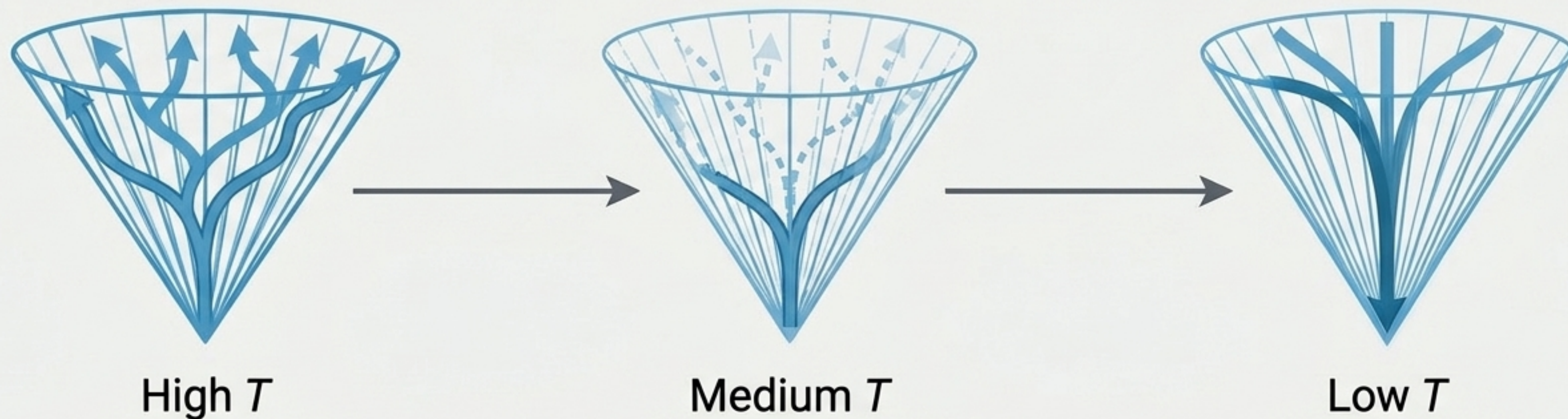
Raising a penalty does not move the ground state; it inflates barriers between states.

Invalidity is an occupied phase of search, not an absence.  
Search manufactures adjacency.



# Annealing as Scheduled Loss

## Scheduled Loss of Alternatives

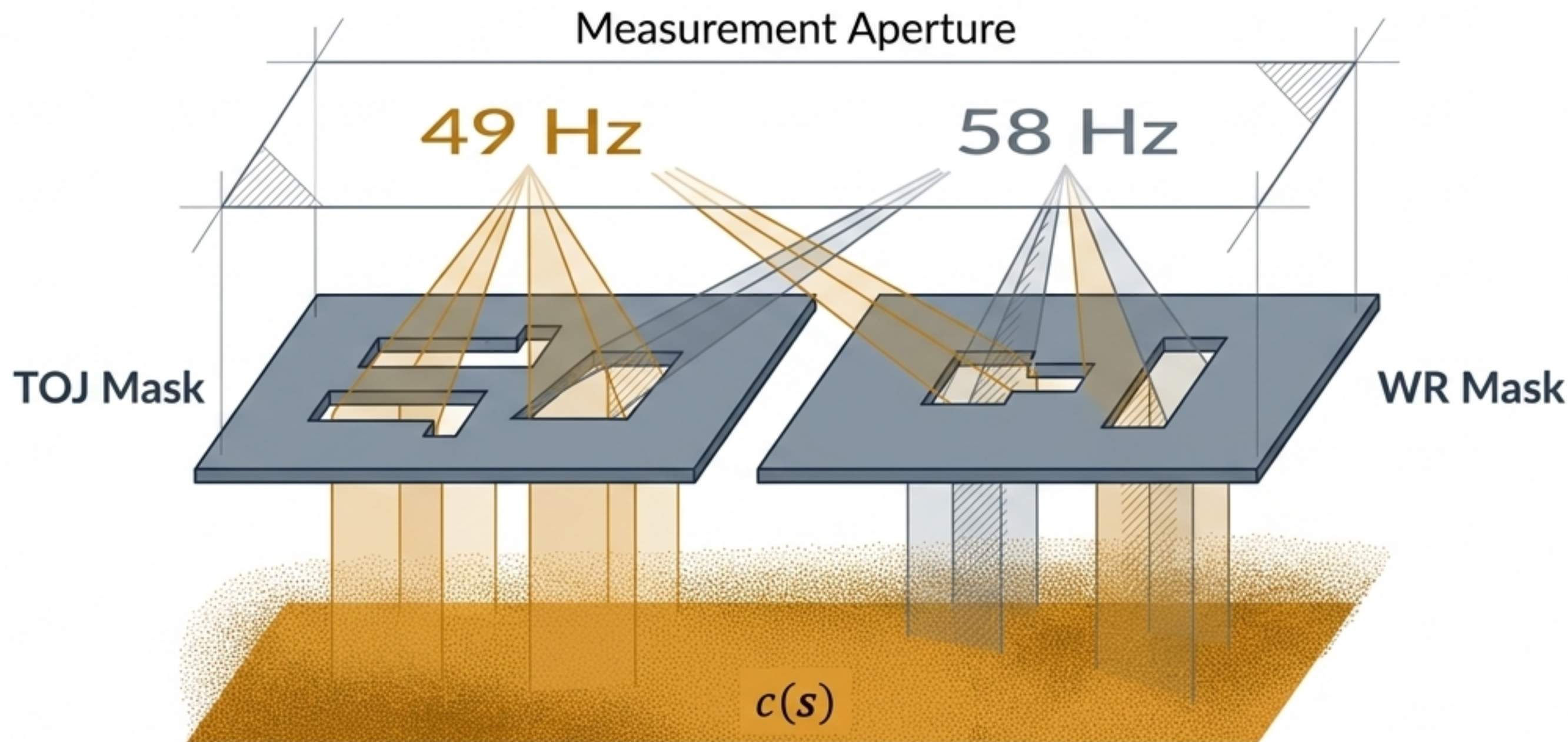


**Axiom 1:** Uphill continuation sets nest as  $T$  falls; downhill sets grow.

Annealing is not simply finding low places. It is the precise scheduling of when possibilities stop being possibilities.



## Case 2: The Task Enters the Quantity

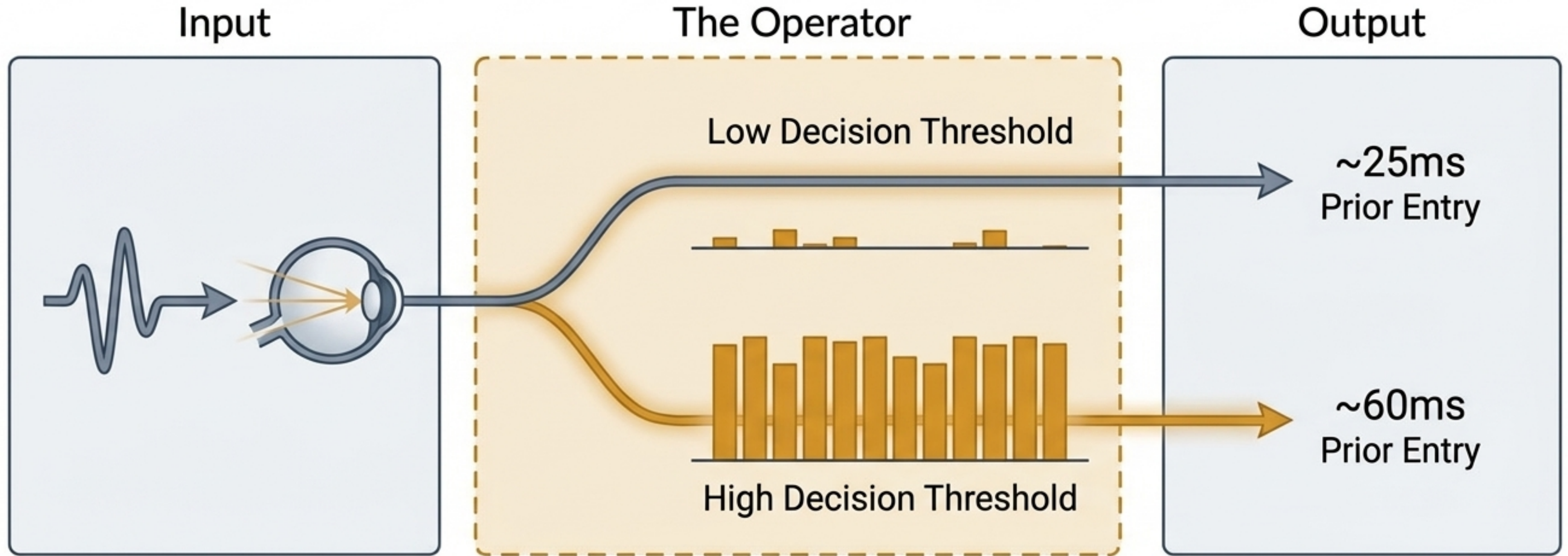


A state  $z$  does not determine its measurement.  
When testing Visual Short-Term Memory, a nominally task-independent parameter does not transport cleanly across tasks.



# Invariance is Relative to a Transformation

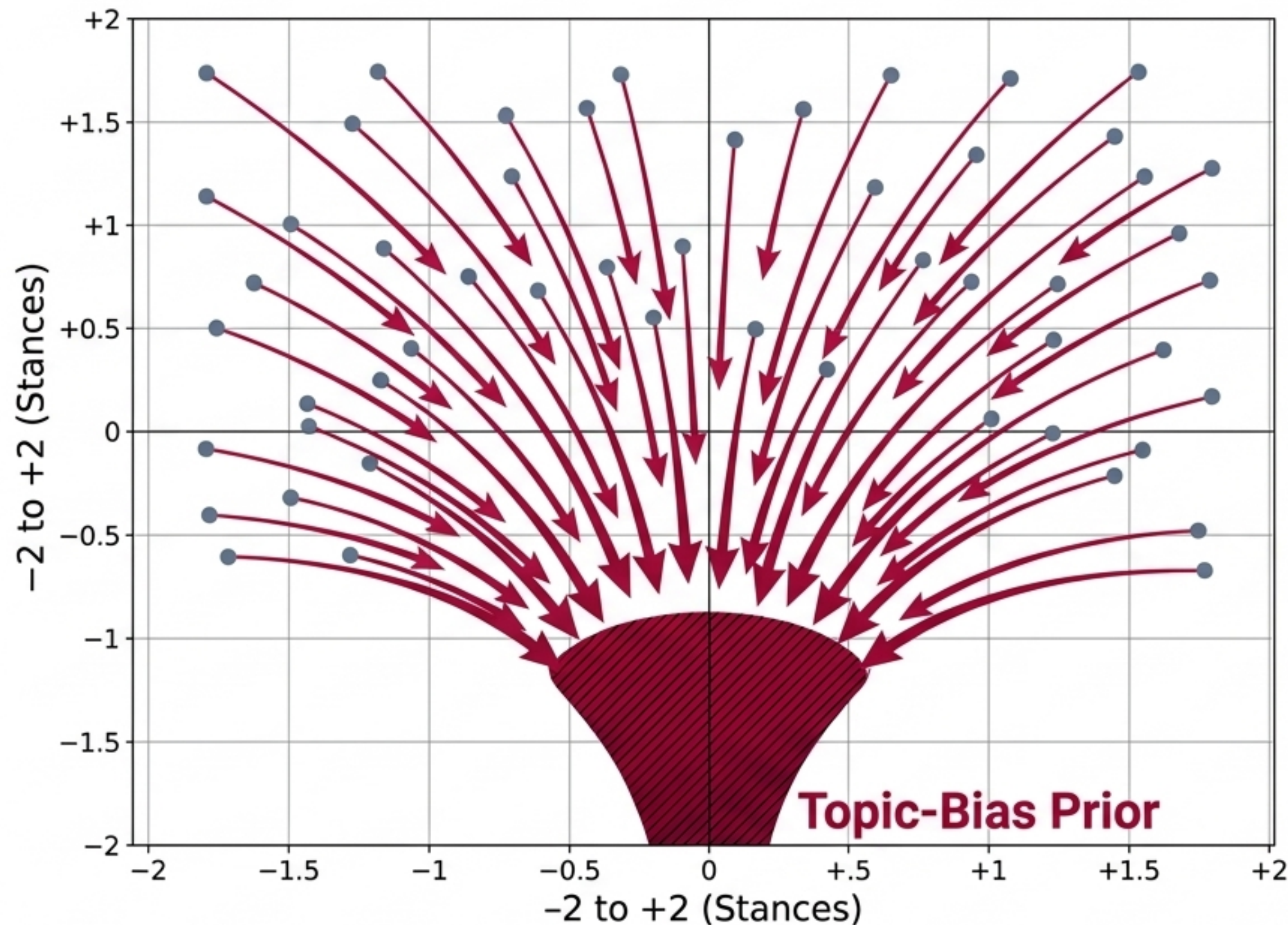
$$\hat{C}^{(a)} \neq \hat{C}^{(b)} \Rightarrow C_a^* \neq C_b^*$$



A one-parameter model acts as a drain for task variance. Difference in the output estimates does not prove the latent capacities are different—it proves the measurement operators are.



# Case 3: Shared Destination, Separate Causes



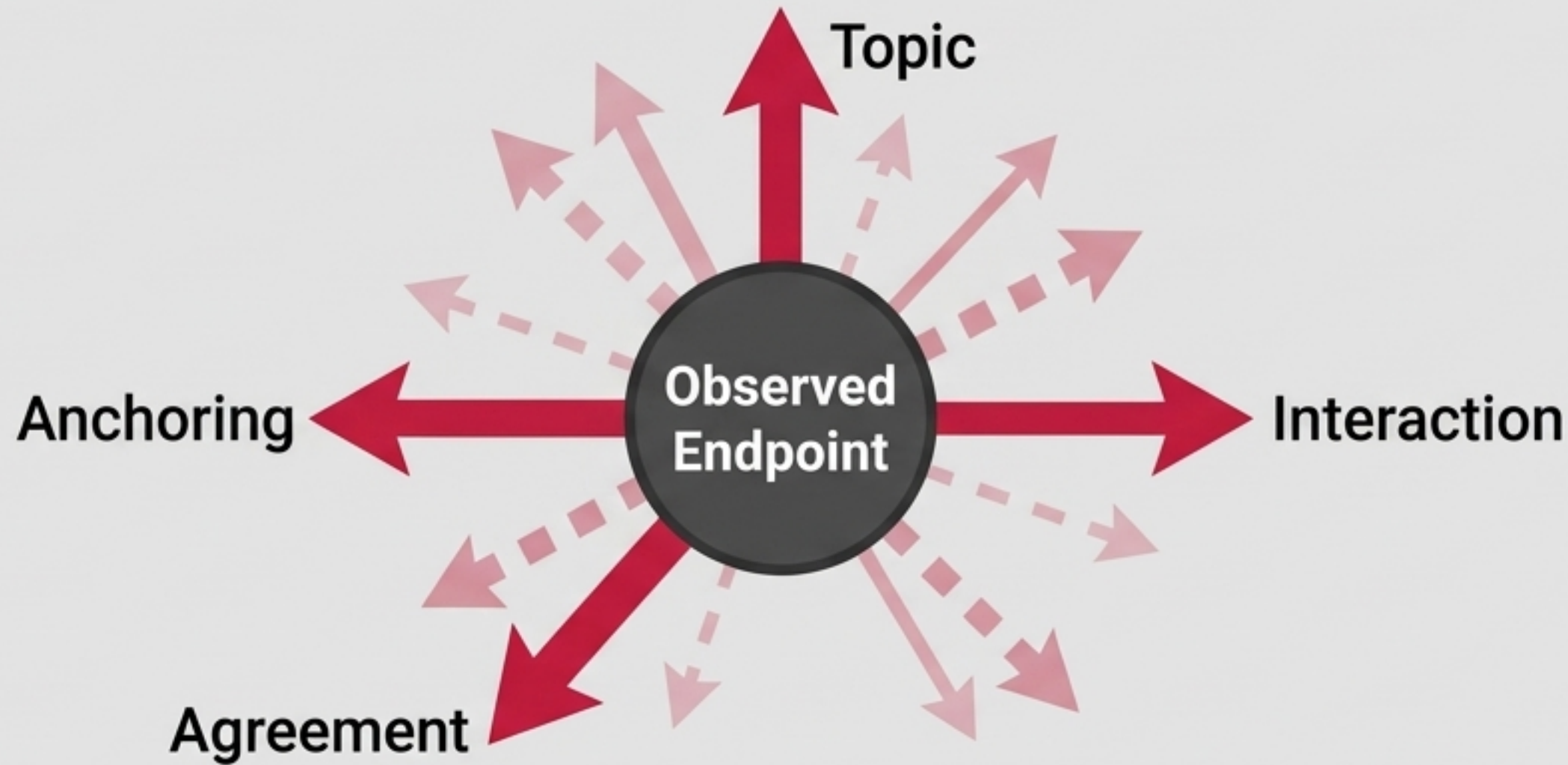
**Co-motion  $\neq$  coupling.**

When Language Models engage in **dialogue and converge on a stance**, it is rarely persuasion.

Agents instantiated from the same model begin the conversation **already sharing a geometry.**



# Breaking the Degeneracy



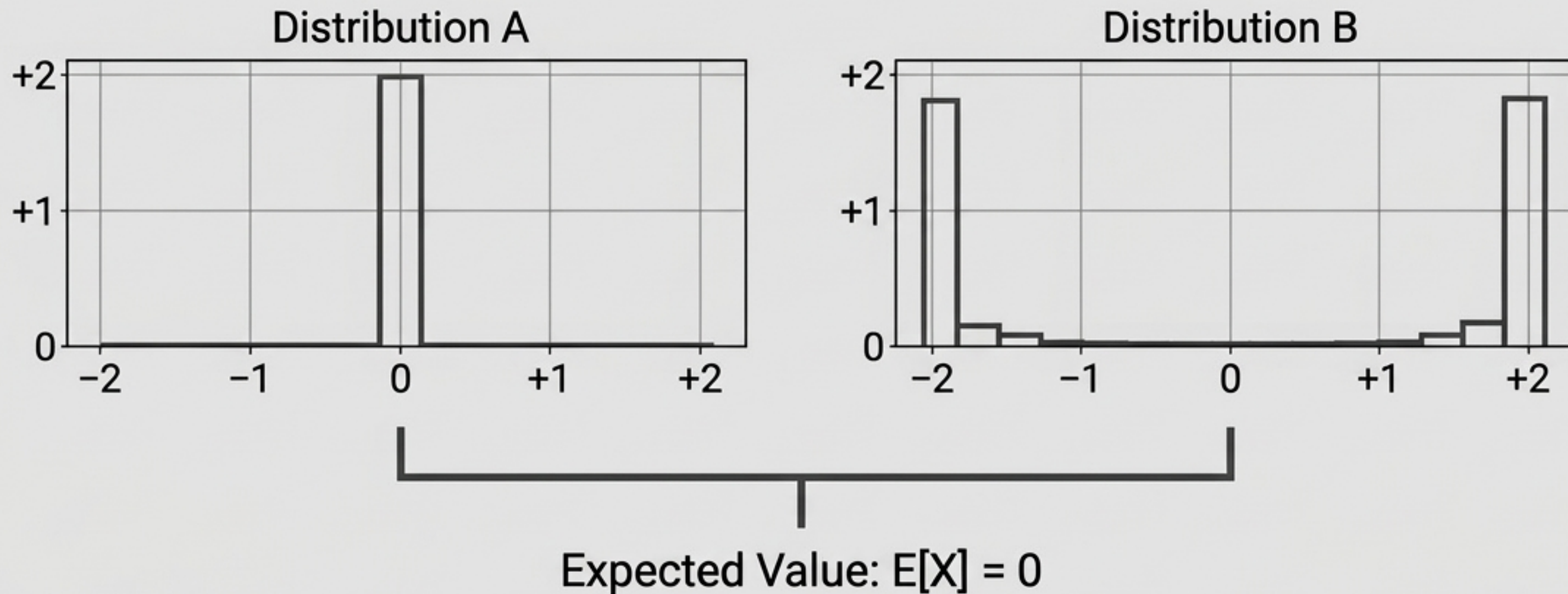
**Axiom 4:** Linear pulls reduce to a centroid;  
the endpoint identifies at most the centroid.

You cannot separate the causes of a trajectory just by watching where it ends.  
A shared destination does not imply a shared cause.



# Uncertainty Has Shape

$E[X] = 0$  does not determine  $H(X)$ .



The expected value discards the distribution. A sharply concentrated neutral agent and a diffusely uncertain neutral agent occupy the exact same coordinate, but have radically different continuation structures.

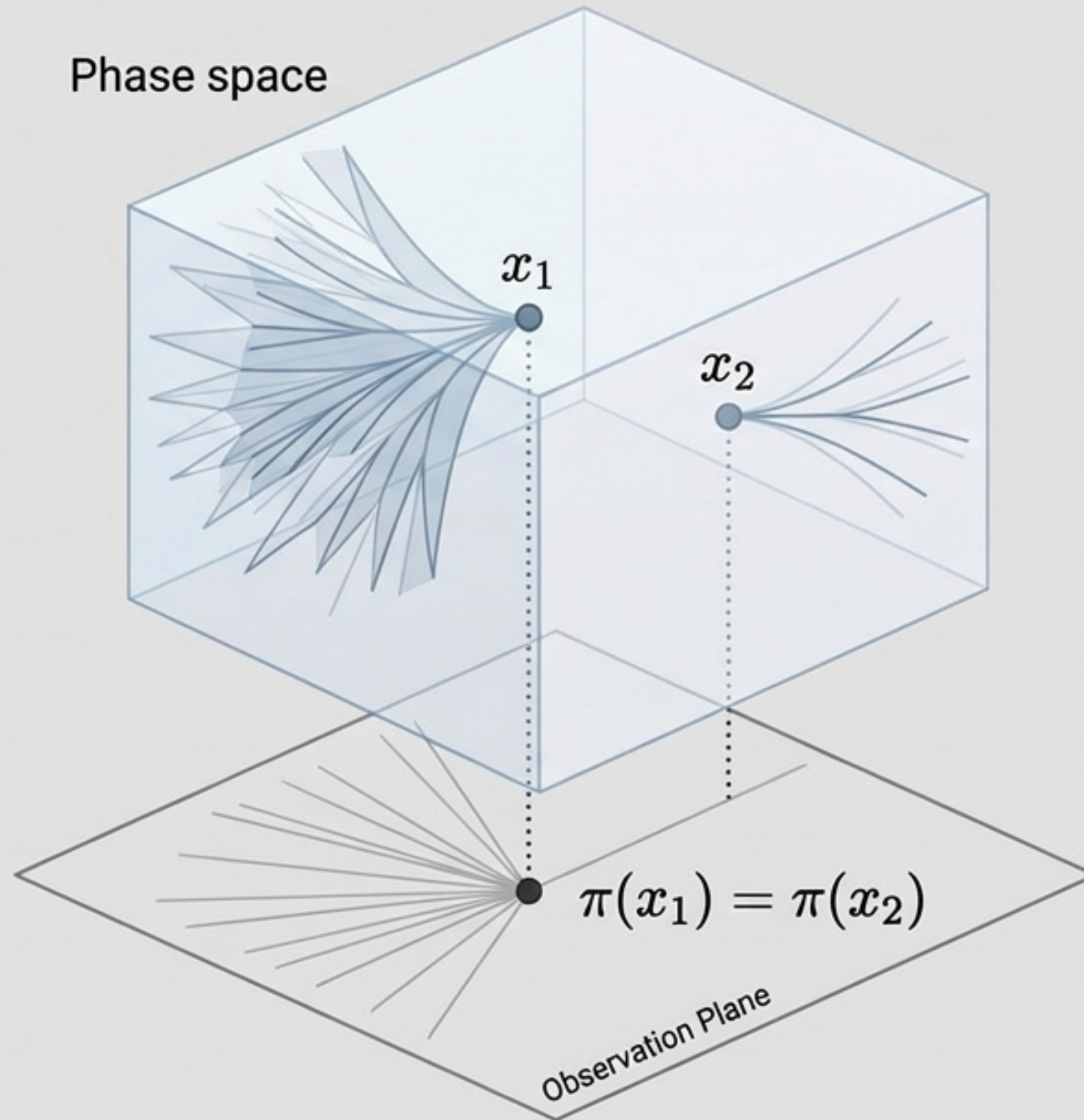


# The Three Missing Terms

|                     | Physical<br>(Ising)             | Cognitive<br>(TVA)           | Conversational<br>(LLM)        |
|---------------------|---------------------------------|------------------------------|--------------------------------|
| The Illusion        | Endpoint reveals<br>the minimum | Endpoint reveals<br>capacity | Endpoint reveals<br>persuasion |
| The Missing<br>Term | K (Kernel)                      | M (Measurement)              | L (Attractor)                  |
| The Reality         | Matter computes<br>probability  | Operator absorbs<br>variance | Prior dictates<br>destination  |



# Grades of Equivalence



## Quotient Geometry

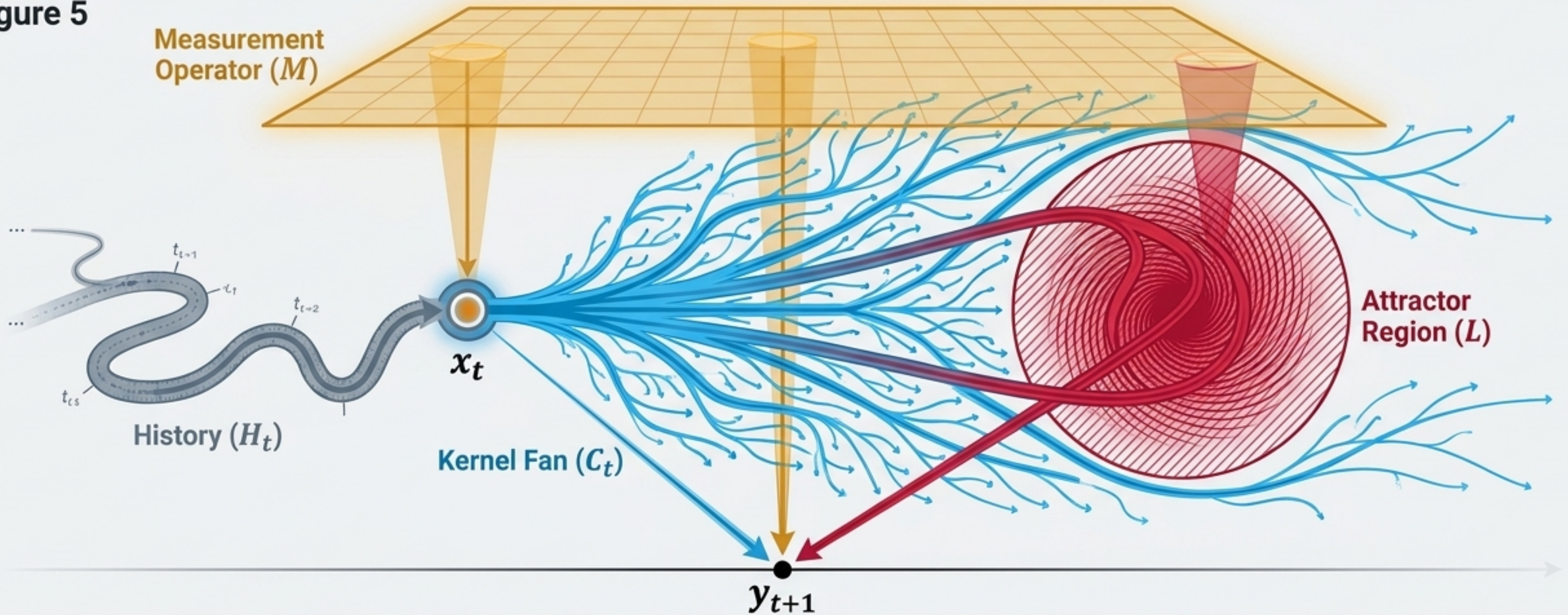
State descriptions are lossy quotients. A shared label establishes equivalence under the labeling map and nothing more.

A compression is admissible only relative to the future distinctions that must survive it.



# The Composition

Figure 5



$y_t = M(K^t(x_0))$  A state is intelligible only relative to the dynamics that reach it, the measurement that reveals it, and the continuations that remain from it.